World Studies Extended Essay

Economics and Biology

Health and Development: Global Health Crisis

Cost-Effectiveness of Flu-Vaccination: A Case Study of China and England

Research Question: Given the risks of influenza, both annual epidemic and pandemic: How cost-effective would it be to expand routine influenza-vaccination universally to children and adults of all ages?

3973 words

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1.0 INTRODUCTION

Seasonal influenza (flu) is a costly disease both in terms of medical-cost and lost productivity[1]. Belonging to the family *Orthomyxoviridae,* Influenza is a non-cellular infectious virus consisting mainly of Ribonucleic-Acid enclosed by a lipoprotein envelope[2]. Of the four types of Influenza: A, B, C, and D, only A, B, and C affect humans. Due to its ubiquity and contagiousness, it accounts for an estimated annual global mortality of 250,000 to 500,000, especially among persons with high-risk comorbidities/groups (HRGs) – notably infants and seniors.[3]

Among the fastest ageing populations[4], China's and England's demographic-shifts present considerable challenges both in terms of the epidemiology of viral-infections and the cost-effectiveness of its control. While implementing routine vaccination for the wider population is an effective defence against flu outbreaks through the creation of herd-immunity[5], budget-constraints limit the ability of countries to actualize such broad immunization-programs.

This World Studies Extended-Essay examines the question whether implementing a universal vaccination-program is more cost-effective than the current targeted vaccination of HRGs around the world, using China and England as case studies. The analysis rests on an interdisciplinary approach combining necessary disciplines from Economics and Biology which enables socio-economic policy choices that deliver health outcomes cost-efficiently. It draws on existing literatures: professional journals, E-books/books, websites,

[1] Duncan G. Ian et al., "Planning influenza vaccination programs: a cost benefit model. Cost Eff Resour Alloc" (26 July 2010), Research Article: no.10, [Accessed June 02, 2020].
[2] Wagner R. Robert, *Virus,* (Department of microbiology, University of Virginia), 26.
[3] Paget John, et al., "Global mortality associated with seasonal influenza" (World Health Organisation ,J Glob Health, 09 December 2019), [Accessed June 02, 2020].
[4] China Power Team, *Does China have an ageing problem,* (China Power, 15 Feburary 2016),1. Website. [Accessed June 03, 2020].
[5] Sadarangani Manish, *Herd immunity: How does it work?,* (Oxford Vaccine Group, 26 April 2016). Website. https://www.ovg.ox.ac.uk/news/herd-immunityhow-d
oes-it-work. [Accessed July 7, 2020]

videos, economic and epidemiological models such as Preaud-et-al (2014), Duncan-et-al (2010), and Ting-et-al (2017), selected for their reliable, meticulously scrutinized investigations using statistics from professional-standard sources.

The analyses found that universal vaccination is more cost-effective than targeted vaccination in both England and China in the long-term (Vaccinating Seniors only was more cost-effective than vaccinating infants only). Vaccinating HRGs only was more cost-effective in England than in China. Consistent with existing literature, it concludes that seniors are more vulnerable to Influenza like illnesses (ILIs) than young adults, and that universal vaccination delivers greater health-benefits due to herd-immunity induced by vaccines.

1.1 The Epidemiology of Influenza

Influenza-A and B viruses are the only Influenza-viruses known to cause disease pandemics or epidemics. A pandemic can occur when a new and very different influenza-A virus emerges that infects people, and has the ability to spread efficiently between people.[6] Such viruses can be transmitted up to 6ft away mainly via droplets made when an infected person coughs, sneezes, or talks. They can survive on dry surfaces up to three days, and on fabrics eight to twelve hours post-contamination.

Most common, Type-A can be subclassified based on two of the glycoproetins on its protective surface through serologic testing; Hemagglutinin (H) which binds to host cell-membranes, and Neuraminidase (N) which are enzymes required for viral-replication used to cleave sialic-acid groups. Eighteen and eleven different H and N subtypes respectively mean a potential 198 different viral-versions, of which 131 have been identified in nature.

Influenza-B viruses are separated into two genetic lineages: B/Yamagata and B/Victoria on the basis of their hemaglutinin glycoproteins. Type-B viruses' genetic and antigenic properties change slower than Type-A's - especially A(H3N2) viruses. Nonetheless, both have been responsible for infections since the 1980s. Types-A and B can be further subdivided into different genetic-clades and sub-clades based on similarities in their hemaglutinin-gene sequences. This allows specialist-tracking proportion of viruses from different clades in circulataion. Clades and sub-clades are genetically different, but not necessarily antigenically different - they impact host-immunity similarly.

[6] *Types of Influenza viruses,* (18 November 2018). Website. https://www.cdc.gov/flu/about/viruses/types.htm, [Accessed July 15, 2020].

Type-C has only one type of glycoprotein, Hemagglutinin-Esterase-Fusion, which it uses to enter and exit cells. This Subtype of the Influenza-virus can only infect humans and pigs, and is the least likely of the three to mutate.

Epidemiologically, Influenza outbreaks are seasonal, often sporadically occuring during early spring, late fall, and winter seasons. Influenza A(H1N1), A(H3N2), and B-viruses which cause disease in humans, often co-circulate globally. New variants can emerge as a result of point mutations and recombination events that occur during viral-replication, resulting in frequent antigenic-shift or antigenic-drift; this serves as the virologic basis for seasonal epidemics. This makes adjustment of vaccines for Influenza-viruses each season neccessary, which could become cost-innefficient over time.

Influenza-virus normally has a high morbidity-rate, but may have low-case mortality (usually 0.003% to 0.03%)[7]. However, some such as Avian-flu, is caused by highly-pathogenic Influenza-viruses that can cause case fatality up to 60%[8]. Approximately 9% of the world's population is affected annually, with up to 1billion infections; 3 to 5 million severe cases; and up to 500,000 deaths each year[9]. Since the 1970s, clades of Influenza A(H3N2) and A(H1N1) have been the most common viral-subtypes affecting humans and few mamals. These were responsible for the Spanish, Asian, Hong Kong, and Swine Flus which occurred in 1918, 1957, 1969 and 2009 respectvely. Of these, the Spannish flu was the most deadly[10], with an estimated mortality between 20–50 million (Table 1)[11]. Historically, rates of serious illness and deaths are highest amongst persons aged 65 years and above, children under 2 (occaisonally+/-10) years, and persons with ILIs and underlying health contitions.

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[7] Zhong Nan-shan, et al, *Journa of Thoracic Disease,* (Influenza Diagnosis and Treatment Expert Panel of the Chinese Ministry of Health, December 2011), vol 3, 4.

[9] Clayville R. Lisa, "Influenza Update: A Review of Currently Available Vaccines*",* (Pharmercy and Therapeutics, October 2011) Vol 3, 10.

[10] NIAID, *What was the 1918 Influenza Pandemic?,* video, https://www.youtube.com/watch?v=L6jTMacxzkI, [accessed July 8, 2020].

[11] Henehan Carl,Tom Jefferson, "Covid-19 deaths compared with "Swine flu*"",* (09 April 2020). Journal. [Accessed August 21, 2020].

Table 1.0: Key Incidences of Influenza-A and Mortality Rates:[12]

Influenza A subtype	Strain	Year	Approx. people infected worldwide	Approx. estimation of death toll worldwide
Spanish Flu	H1N1	Jan 1918 – Dec 1920	500mil	20 mil to 50 mil
Asian Flu	H2N2	1957	N/A	1.1 million
Hong Kong Flu	H3N2	1969 – 1970	N/A	1.0 million
Avian Flu	H5N1	1997	399	251
Swine Flu	H1N1	Jan 2009 – Aug 2010	N/A	284.5 thousand

In China, 115 and 142 people per 100,000 popluation with influenza-related severe acute respiratory infections (SARI) were hospitalized between 2010-11, and 2011-12, respectively.[13] In addition, morbidity-rates due other ILIs was estimated at 8.8 and 12.4 per 100,000 in China's southern and northern regions, respectively.[14] More recently, with the advent of COVID-19, the rate of morbidity and mortality due to viral-infections has increased significantly around the world.

1.2 Current Methods of Influenza Flu Prevention, Treatment and Costs

There is a concensus among Epidemeologists that universal vaccination is the best option to prevent/control Influenza, and decrease its socioeconomic burden[15]. Given the genetic variability of Influenza-viruses, finding cost-effective prevention and treatment continues to be a top priotity for

[12] *Pandemic* Influenza: Past pandemic, (euro.who.int). https://www.euro.who.int/en/health-topics/communicable-diseases/influenza/pandemic-influenza/past-pandemics. Website. [Accessed July 08, 2020].
[13] Huang H. Yu, Huai Y. J., "The substantial hospitalization burden of influenza in central China: surveillance for severe, acute respiratory infection, and influenza viruses", *(* Influenza Other Respir Viruses, 08 January 14). Journal.
[14] "Influenza-associated mortality in temperate and subtropical Chinese cities, 2003-2008*"*, (Bull World Health Organ. 01 April 2012). PMC Article, https://pubmed.ncbi.nlm.nih.gov/22511824/, [Accessed October 18, 2020]
[15] Clayville R. Lisa, "Influenza Update: A Review of Currently Available Vaccines*"*, (Pharmacy and Therapeutics, October 2011), Journal Article. vol. 36, 10.

medical research.[16] Specifically, understanding the cost-effectiveness of different vaccination-programs (universal or targetted) is crucial for delivering health-benefits at minimal cost, especially in countries with limited fiscal-capacity.[17]

The World Health Organisation's strategic Advisory Group of Experts on Immunization recommends a targetted annual seasonal Influenza-vaccination be administered to HRGs.[18] Notwithstanding, China does not currently have a national government-funded seasonal flu-immunization program so uptake is generally low. (Seasonal Influenza-vaccine coverage rate was 1.9% of the population).[19] Conversely, England enjoys robust National Health Services (NHS) that provide flu-vaccinations for HRGs.

1.2.1 How vaccines work against influenza

Primary and seconadary barriers to disease causing pathogens are a cruial part of our defence-mechanism. These include skin, mucous-membranes, eyelashes, cillia-cells, stomach-acid, leucoplasts, and complement-proteins responsible for the initiation of the immune-response. Approximately fifteen-quintillion of thrity types of these complement proteins in our bodies remain passive until activated. In face of infection, the complemnt-cascade activates through one of three pathways. (Figure 1).

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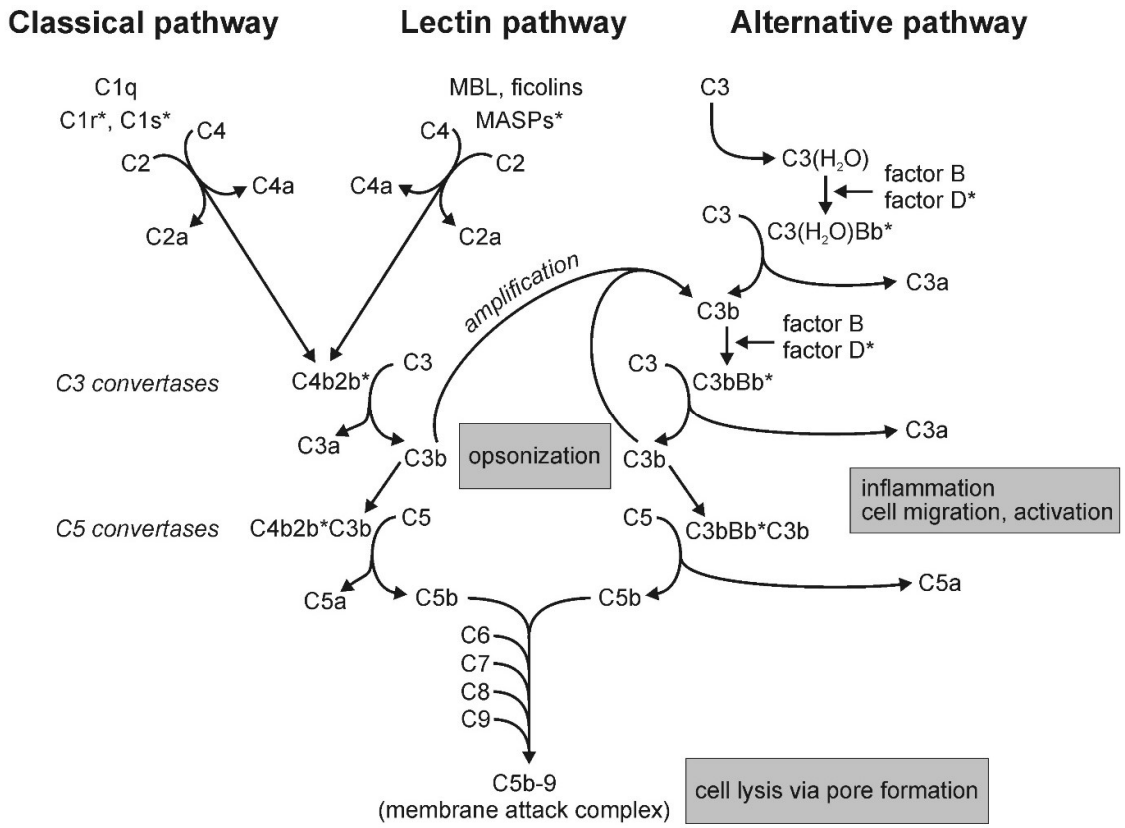
[16] Zhong Nan-shan, Li Yi-min, Yang Zi-feng et al, "Journal of Thoracic Disease", (Influenza Diagnosis and Treatment Expert Panel of the Chinese Ministry of Health, December 2011), PMC Article, vol 3, 4.

[17] *The economic burden of influenza-associated outpatient visits and hospitalizations in China: a retrospective survey*, (ncbi.nkm.nih.gov, 06 October 2015). Website. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4595124/, [Accessed October 19, 2020].

[18] *Vaccines against influenza WHO position paper* – (Wkly Epidemiol Rec., 23 November 2012). E book, 87(47):pg.461-476.

[19] Feng L, et al, "Vaccine,Seasonal influenza vaccine supply and target vaccinated population in China, 2004-2009", (24 September 2010). Research Paper, 28(41):6778-82.

Figure 1.0: Diagram showing simplified workings of the complement-system:[20]



[20] Mgr. Pastorek Michael, PhD, *Immune disorders,* (Institute of Molecular Biomedicine (IMBM)). Academic paper.

Viral-reproduction requires transition to new cells. Outside cells, they are vulnerable to complement-MBL which intercepts and cripples them before the infection spreads. For weak viruses, inclusive of Influenza-C, viral-infection can terminate before symptoms appear. However, selective-pressures mean viruses also evolve mechanisms to bypass the immune-system. For example, when the virus *Vaccinia poxvirus* infects a cell, it forces the host-cell to produce a protein that prevents compliment activation. This creates safe-zones around the infected cells, allowing further infection. If pathogens should withstand the complement-system and become life-threatening, intelligence dendritic-cells will begin to gather information about the pathogen and activate B and T-lymphocytes which produce antibodies that bind to specific epitopes on pathogenic-antigens and destroy them. The process takes time, and inefficiency will result in death. To counter this, memory-cells are produced which remain passive until a similar pathogen reappears; immune-response would be instantaneous and effective.

Everyone is born with this elemental innate-immunity, but adaptive-imunity develops throughout our lives by a person's own immune-system after exposure to an antigen through a disease or an immunization (Table 2.0).[21]

Table 2.0: Types of Immunity

Type of Immunity	Process
Natural Active-Immunity	Develops from primary-immune-response when exposed to a live pathogen post disease contraction.
Natural-Passive Immunity	Short-term immunity passed on from a mother to a child through the placenta or breast milk.
Artificial Active-Immunity	Develops when injected with a safe (weakened or dead) form of the pathogen which can be easily subdued by the immune system to produce memory-cells without serious risk.
Artificial-Passive Immunity	An immediate, short-term immunization provided by the injection of antibodies that are not produced by the recipient's cells.

[21] *Immunology basics What is Immunity?*, (Health.mil; and *Immunity Types,* 10 March 2017). Website. [Accessed November 08, 2020].

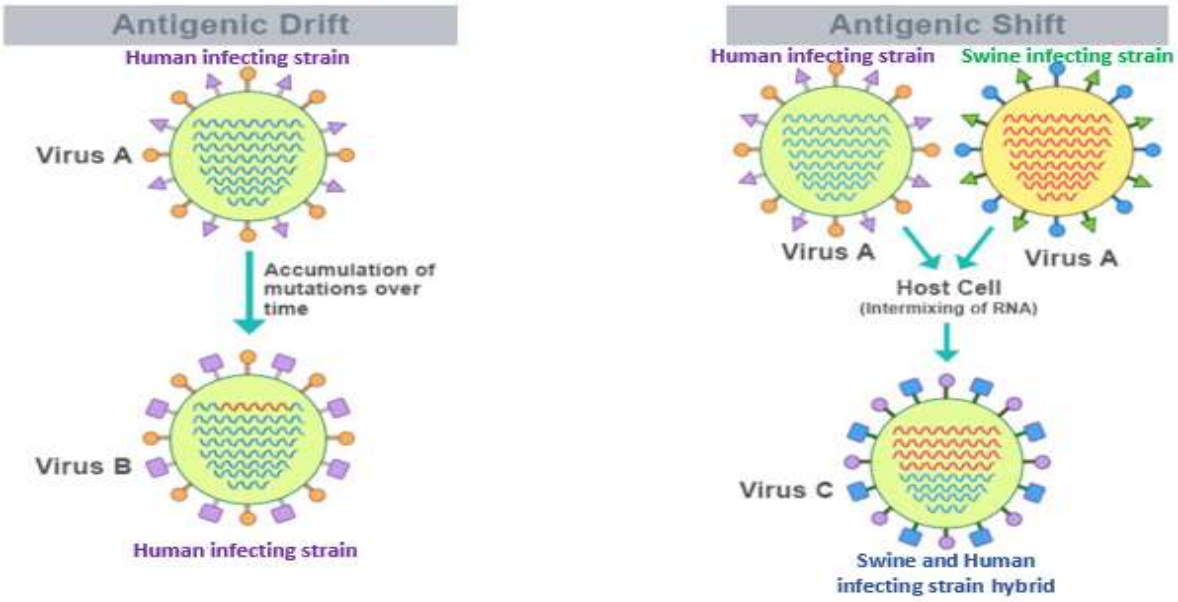
Gaining memory cells via infection is an unpleasant and dangerous method of gaining natural-immunity, so vaccines are essential. The seasonal-flu vaccines are revised and designed annually to provide protection against three or four strains that may spread within upcoming seasons by provoking immune-response and thus providing the necessary memory cells. The vaccines are biological-preparations that provide artificial-immunity to Influenza-viruses, usually containing one weakened A(H1N1), A(H3N2), and Influenza-B virus (trivalent). Strain growth mostly takes place inside chicken eggs, but can also be produced inside mammalian cells or recombinant technology.[22]

1.2.2 Problems with Vaccination: Antigenic-drift and Antigenic-shift

Influenzas A and B have high tendancies to mutate their hemaggultinin and neuraminidase glycoproteins during reproduction, and are therefore most prone to cause severe illnesses. Viral-mutation occurs when genetic-code is altered by change to a ribonucleic-acid gene. Because a protein's function is determined by its shape, and its shape by genetic-code, mutation means the process of gaining artificial/ natural immunity must be repeated to counter said pathogen in face of reinfection with new memory-cells. Antigenic-drift occurs due to variation in the viral-genome by a accumulated mutations over time. Antigenic-shift is an abrupt. It's major change brought about by intermixing of genomes from two or more strains of a virus. When two similar viruses infect an animal, RNA might reassort to form new viruses capable of infecting that animal and humans. These viral-mutations occur frequently due to splicing and editing during viral-reproduction. Such changes cause mutations in H and N genes which reduce/inhibit the binding of neutralising antibodies, allowing the virus to bypass immune-response, and are responsible for the Asian flu, Hong Kong flu, Avian flu, Swine flu, and COVID-19.

[22] Teich Daniel, *The challenges of neutralizing Influenza and the quest for a Universal Vaccine*, (Frontiers, 15 December 2017). Journal.

Figure 3.0: Antigenic Shift Vs. Antigentic Drft[23]



Every 2-8 months, a virus that does not match viral-strains in previous vaccines arise resultant from antigenic-drift[24].

[23] HealthHearty, *Antigenic Drift Vs. Antigenic Shift*, General Health. Website, https://healthhearty.com/antigenic-drift-vs-antigenic-shift, [Accessed November 11, 2020]. Edited.

2.0 METHODOLOGY

2.1 Case Studies

The past three centuries have seen six flu-pandemics: four in the 20th century - three originating in China.[24] Dr. Peter Daszak, President of EcoHealth Alliance, referred to South-Central China as "a mixing vessel" for viruses. Densely populated areas like China with regular contact with animals that potentially carry transmissible diseases present high-risk of antigenic-shift. It's estimated that China's population will reach 1.46billion by 2035 with the current death-rate of 9.8 per 1,000 population .[25] On the other hand, England has an aged population, low population density, and a functioning NHS. These present unique samples for comparative evaluations of the cost-effectiveness of universal vs. targeted vaccination around the world.

2.1.1 Research Question

Given the risks of influenza, both annual epidemic and pandemic: How cost-effective would it be to expand routine influenza-vaccination to children and adults of all ages?

Recent literature confirms that influenza-vaccination confers health-benefits and reduces direct and indirect-costs associated with flu in China[26]. The rate of hospitalization and mortality were highest among the elderly. Increasingly, Chinese regional governments have begun to partially/fully fund Influenza-vaccine production and administration for HRGs. As of 2016, basic social medical insurance was provided in 61 regions, 8 provinces, 45 prefectures, and 8 counties. Finance-reimbursed vaccination was offered

[24] Feng L, et al, "Vaccine,Seasonal influenza vaccine supply and target vaccinated population in China, 2004-2009", (24 September 2010). Research Paper, 28(41):6778-82.
[25] *China Population.* Website, https://www.worldometers.info/world-population/china-population/, [Accessed November 21, 2020].
[26] Studies did not examine southern China, a speculative influenza epicentre for the emergance of pandemic influenza-viruses.

mainly for the elderly and school-children in Beijing, Dongli district in Tianjin, Shenzhen, etc. The avarage national vaccination coverage was just 1.5%-2.2% in recent years. A free national vaccination program for priority populations (416million) would cost government US$757,000,000 (4,988,327,200Yuan) with an uptake-rate of 20%.[27] However, the small-scale policy failed to increase national uptake: a free nationwide program would demand substantial annual invesment, which would require cost-effectiveness analyses to maximise efficiency and improve coverage. In England, the recent Coronavirus pandemic has revealed the pressures health-systems face due to ILIs and the need to examine my research question regarding the cost-effectveness targeted vaccination-programs.

2.2 Tools and Methods

Cost-effectiveness analysis (CEA) is a technique for identifying the minimal-cost option to meet a specific outcome. In this case, it is the ratio of the costs and effects of each vaccination scenario.

$$\sum_{s=1}^{n} C_{vp/} E_{vp}$$

C = *The cost items measured in monetary units*
E = *The effectiveness of the Vaccination Program*
s = *The Scenario*
vp = *Vaccination program*

Assumptions (according to WHO)

I) All elderly population (≥60years, and Infants ≤15years) are classified HRGs.

II) All population between the ages of 15–59 years are classified as low-risk-group (LRG).

[27] "Seasonal influenza vaccination in China: Landscape of diverse regional reimbursement policy, and budget impact analysis", (Vaccine, 11 November 2016). Journal, vol34, Issue 47 pages 5724-5735. [Accessed September 29, 2020].

Scenarios

To address the research question, the essay considers three vaccination scenarios in each country:

1) Base scenario: No vaccination.
2) Targeted vaccination: HRGs only. This includes three sub-scenarios: infants only, seniors only, and infants and seniors together.
3) Universal vaccination: (HRG+LRG)

Measurements

a) Cost (measured monetary units).
b) Effectiveness; measure of vaccination-effectiveness (*quality-adjusted-life-year).* QALY combines health-intervention effects on mortality and morbidity into a single measure of effectiveness.

In this essay, I used direct-costs of vaccination and averted deaths as a measure of cost and effectiveness for reasons below.

Researchers have used several models to estimate cost-effectiveness and benefits of flu-vaccines. Preaud-*et-al* (2014)[28] and Franlin et Haschlaf (2018)[29] provided a static-model of vaccination-effectiveness:

$$\sum_{r=1}^{6} \boldsymbol{V_r E_r L_r}$$

V = *The vaccinated coverage-rate*
E = *The efficacy-rate*
I = *The incidence-rate*
r = *The risk groups*

[28] Emmanuelle Preaud, et al, "Annual public health and economic benefits of seasonal influenza vaccination: a European estimate", (BMC public health, 07 August 2014). Journal, 14(1), pg.813.
[29] Franklin Ben and Dean Hochlaf, "An Economic Analysis of Flu Vaccination" (Seqirus, International Longevity Centre UK (ilc), 05 July 2018). Report.

This model would allow estimation of avertable cases through vaccination, depending on coverage, efficacy, and incidence. Duncan-*et-al*[30] developed a model that estimates the costs and benefits of influenza-immunization programs based on published literature and customization of vaccination data by population age and level-of-risk, projection-years, and potential pandemic risk, ideal to estimate net-cost savings per vaccinated individual. Ting-*et-al.*[31] used citation databases and reviews of literature to estimate cost-effectiveness of Influenza immunization-programs. Using data from grey and published literature, they discovered that universal programs provide larger benefits, but value for money remained unclear.

Due to insufficient resources and boundaries of research, my essay adopts a simplified version of the three methodologies: extrapolation of data from literature, scenario analysis, and deductive-reasoning[32] to estimate the cost-effectiveness of flu-vaccination in China and England.

3.0 RESULTS AND DISCUSSION

3.1 Estimated costs and Effectiveness of Flu Vaccination in England and China

To estimate the full costs of the flu requires robust datasets on several variables, including direct-costs and indirect-costs: human capital costs, etc[33]. Data from Ben Franklin & Dean Hochlaf (2018)[34] estimated the costs of vaccine provision by the NHS across GP practices and pharmacies in England. They estimated

[30] Duncan G. Ian, *et al.*, *"Planning influenza vaccination programs: a cost benefit model",* (Cost Eff Resour Alloc 26 July 2010), Article no.10.
[31] Ting E. K. Eon et al., *Systematic review of the cost-effectiveness of influenza immunization programs,* (National Library of Medicine (NIH), 04,April 2017). Website, https://pubmed.ncbi.nlm.nih.gov/28284681/#:~:text=Studies%20of%20vaccinating%20all%20versus,in%20%2435%2C000%20per%20QALY%20gained. [Accessed December 10, 2020].
[32] Bradford Alina, *Deductive reasoning Vs. Inductive reasoning*, (25July 2017). Website, https://www.livescience.com/21569-deduction-vs-induction.html, [Accessed February 8, 2021].
[33] Spielman Ethan, "Direct Costs vs. Indirect Costs: What Are They, and How Are They Different?", (Business News Daily, updated 12 June 2020), Newspaper.
[34] Franklin Ben, Hochlaf Dean, "An Economic analysis of Flu Vaccination*",* (July 2018). Report.

in recent years, the cost of a vaccine dose was £17.13 by GP delivery, and £14.78 when delivered through pharmacies. However, information on the economic-costs of Influenza related diseases and associated outpatient visits and hospitalizations at the national-level in China was not all readily available. I used some germane data from the health-policy study on cost-effectiveness of flu-prevention in Hong Kong conducted throughout a "mild influenza" year[35] which is comparable to data that would apply to China also. The study predicted more than 660,000 ILI cases in a non-epidemic year, with an average ILI-associated cost of 235.47Yuan/case (GBP£27.05) and vaccination-associated costs of 61.57Yuan/vaccinated individual (GBP£7.14).[36] Another case study showed that of the 2,702,226 flu cases reported in mainland China through CNIDRS between 1st/January/05 and 31st/December/18, 55.8% were male and 44.2% were female. The proportions of cases in age groups 0-9, 10-19, 20-59 and 60+ years were 46.9%, 14.8%, 20.8%, and 8.8%, respectively. Yearly incidence rates were 0.35, 0.44, 0.28, 0.32, 1.49, 0.48, 0.49, 0.91, 0.96, 1.59, 1.44, 2.24, 3.31, and 5.51 per 10,000 population from 2005-18 respectively, showing an increase over time.[37]

For both cases, I used the demographic data to calculate the percentage population in different risk categories. I also considered the fact that ahead of pandemic risk, preventive measures such as quarantines and lockdowns are used as public-policy measures to minimize spread, mortality, and maximize economic-welfare. Based on these statistics, I will calculate the cost-effectivenes variables for HRGs (0-15years and 60+years).

[35] Fitzner K. A., Shortridge K. F., McGhee S. M., Hedley A. J., *Cost-effectiveness study on influenza prevention in Hong Kong,* (Health Policy, June 2001), 56(3):215-34.
[36] Fitzner K. A., Shortridge K. F., McGhee S. M., Hedley A. J., Cost-effectiveness study on influenza prevention in Hong Kong, (Health Policy, June 2001), 56(3):215-34.
[37] Zhang Yewu, et al, "Spatiotemporal Analysis of Influenza in China, 2005–2018*",* (Scientific reports, 23 December 2019). Report. [Accessed December 11, 2020]

3.1.1 Calculating Vaccination Effectiveness – Averted Death

The estimation of averted cases due to vaccination-programs will be extremely sensitive to changes in coverage-rates, efficacy of vaccines and incidence-rates of the flu in the population. Tthe efficacy of the vaccine is therefore measured in terms of cost per QALY[38]. For this essay, I have used averted deaths as a measure of effectiveness due to deficient data.

The paper examined two distinct scenarios, with the aim of highlighting if it is beneficial to implement targeted vaccination of HRGs only or a universal vaccination program (HRGs+LRGs) to improve health outcomes and cost-effectiveness.

[38] *Quality-Adjusted Life Year* (York, York Health Economics Consortium, (QALY) [online]. 2016). Website, https://yhec.co.uk/glossary/quality-adjusted-life-year-qaly/, [Accessed February 8, 2020].

Table 3.0: Vaccination of Different Age Groups up to 75% coverage rate

Variables	England				China			
	Unit Costs	**Infants under 15**	**60 and above**	**All**	**Unit Costs**	**Infants under 15**	**60 and above**	**All**
Population		10,635,668	10,075,896	55,977,200		256,591,301	260,915,873	1,441,524,160
Mortality Rate of Unvaccinated people		0.03% - 60%	0.03% -60%	0.03% - 60%		0.03 – 60%	0.03 – 60%	0.03-60%
Effectiveness Measure – Averted Deaths		118	8,834	9,872		2.38	1,398	21,819
Cost of vaccine delivery per person through Pharmacy (based on 2014-15 to emulate the cost of a potential 'new' flu virus)	**£14.78**	£117,896379.80	£111,691,307.20	£620,507262.00	**£7.18**	£1,381,744,153.00	£1,405,031,976.00	£7,762,607,602.00
Incidence rates (%)		19.1	4.91	6.42		0.0016	0.03	0.551
Estimated Population vaccinated who had initially got infected		1523559.441	371044.870	2695302.180		3079.096	58706.071	5957098.591
Coverage Rate (%)		75				75		
ALL DATA IN THIS TABLE HAS BEEN PROPERLY REFFERED TO IN THIS PAPER								

*Average calculated from existing literature[1].

England population=55,977,200. 19%<15 years, 18%>60 years[39].

China=1,441,524,160, 17.8%<15 years, 18.1%>60 years

Assumption: Vaccines have the same effectiveness-rate in both England and China subject to variations in demographic data and medical pre-conditions.

[39] Clark D., *Population of the United Kingdom in 2019, by age group,* (Statista.com, 12 January 2021). Website, https://www.statista.com/statistics/281174/uk-population-by-age/, [Accessed September 29, 2020].

3.2 Cost-Effectiveness of Universal Vaccination vs. Targeted Vaccination

Scenario Calculations: Cost-effectiveness of the different scenarios:

1) **Base scenario:**
   No vaccinations.

2) **Targeted vaccination:**

   i) Children only:

$$\text{England} = \frac{£117,896,379.80}{118}$$

$$= £999,121.86 \text{ per death averted}$$

$$\text{China} = \frac{£1,381,744,153.00}{2.38}$$

$$= £3,288,551.08 \text{ per death averted}$$

   ii) Seniors only:

$$\text{England} = \frac{£111,691,307.20}{118834}$$

$$= £12,643.45 \text{ per death averted}$$

$$\text{China} = \frac{£1,405,031,976.00}{1398}$$

$$= £1,005,030.03 \text{ per death averted}$$

   iii) Children and Seniors:

$$\text{England} = \frac{(£117,896,379.80 + £111,691,307.20)}{118+8834}$$

$$= \frac{229587684}{8952}$$

$$= £13,169.84 \text{ per death averted}$$

$$\text{China} = \frac{(£1,381,744,153.00 + £1,405,031,976.00)}{2.38+1398}$$

$$= \frac{2,86,776,129}{1400.38}$$

$$= £1,990,014.23 \text{ per death averted}$$

3) Universal Vaccination

$$\text{England} = \frac{£620{,}507262.00}{9872}$$

$$= £62{,}855.27 \text{ per death averted}$$

$$\text{China} = \frac{£7{,}762{,}607{,}602.00}{21818}$$

$$= £355789.15 \text{ per death averted}$$

According to the cost-effectiveness analysis figures, vaccinating children under 15 only will require £999,121.86 per death averted in England and £3,288,551.08 in China. Exclusive vaccination of seniors over 60 will require £12,643.45 and £1,005,030.03 per death averted in England and China reciprocally; and vaccinating both children and seniors altogether will lead to £13,169.84 and £1,990,014.23 per death averted in England and China, respectively.

Universal vaccination of both the high and low risk populations in England and China will lead to £62,855.27 and £355,789.15 per death averted, respective; it is more cost-effective to vaccinate Seniors only in both countries compared to vaccinating children only. Vaccinating the whole HRG would be more cost-effective in England than in China.

However, universal vaccination would be more cost effective in both England and China than vaccinating infants only. These results are consistent with the literature which confirms that seniors are more vulnerable to high rates of mortality during to ILIs than infants and young adults, and universal vaccination delivers greater health benefits due to herd-immunity induced by vaccines.

4.0 CONCLUSION

Given the risks of influenza, both annual epidemic and pandemic, the essay examined how cost-effective would it be to expand routine influenza-vaccination to children and adults of all ages in China and England. *Orthomyxoviridae* Influenza*s* are prone to cause health epidemics and pandemics.

The analyses found that universal vaccination is more cost-effective than targeted vaccination in both England and China in the long-term. However, vaccinating Seniors only is more cost-effective than vaccinating infants only. Vaccinating the high co-morbidity population (both infants and seniors together) is more cost-effective in England than in China. These results are consistent with the literature which confirms that seniors are more vulnerable to high rates of mortality due to Influenza like illnesses (ILIs) than infants and young adults; and universal vaccination delivers greater health benefits due to herd-immunity induced by vaccines. In conclusion, it is more cost-effective to expand routine influenza-vaccination to children and adults of all ages globally.

My investigation used reliable secondary data from published literature. The analyis would have benefited from primary data, but this was not possible due to time and resource constraints. Additionally, the difference in cost-effectiveness results in England and China could be further explained by socio-economic and heath-systems disparities. However, space limitations constricted further details in this essay. Finally, It used China and England as case studies. I would be interested to expand the analyses to other countries to improve my results in a global context.

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